# Assignment: Building and Fine-Tuning Your Own LLM

**Training Base Models, Instruction Tuning, and Inference**

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# 1 Introduction

This assignment can be done in group and will be graded for extra credits. This assignment is worth the same as a regular class assignment. Make a presentation to address the questions below.

In this assignment, you will train your own *base* Large Language Model (LLM), turn it into an *instruction-following assistant*, and compare it with a stronger pretrained model.
This assignment is **hands-on** and is designed to help you understand:

- How LLMs are trained
- How datasets affect model behavior
- What "instruction tuning" and "assistant models" really mean
- How inference works
- How sampling affects predictions
- The differences between base and aligned models

You will use **Google Colab Pro** (student accounts available) and work with models small enough to run on a Colab GPU.

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# 2 1. Colab Setup

1. Sign up for **Google Colab Pro** using your student account.
2. Create a new notebook in Colab.
3. Choose the **strongest GPU available** (A100 if possible; otherwise L4 or T4).
4. Take a screenshot of:
    - Your Colab runtime settings

- GPU type assigned

**Write a short paragraph** explaining why GPU hardware matters for LLM training.

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# 3 2. Train a Small Base Model on Your Own Dataset

You will train a small transformer (1–10M parameters) using your own text dataset from Google Drive.

## 3.1 2.1 Dataset Requirements

In your presentation, include:

- A preview of the text dataset
- How you obtained the dataset
- Domain of the dataset (e.g., finance, literature, conversations, sports, medical, social media, etc.)
- Dataset size measured in:
  - tokens
  - words
  - file size (KB/MB)

## 3.2 2.2 Model Architecture Requirements

Show the following:

- Vocabulary size
- Context window (block size)
- Embedding dimension

- Number of layers
- Total number of parameters

### 3.3 2.3 Training

- Train the model for enough iterations to observe improvement.
- Save and show your **training loss curve**.
- Report:
  - Total training time
  - GPU type
  - Batch size

### 3.4 2.4 Inference Demo

Show at least **3 examples** of inference:

- Text continuation
- A question about your dataset
- A creative prompt (story, summary, etc.)

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## 4 3. Convert Your Base Model Into an INSTRUCT Model

You will now make your model behave like an assistant.

Use a small instruction dataset such as:

- Alpaca
- Dolly

- OASST
- A small custom dataset you create (Q/A pairs)

### 4.1 3.1 Show the Instruction Dataset

Include:

- Sample instruction → response pairs
- Dataset size (# of examples)

### 4.2 3.2 Train an Instruction-Tuned Version

Use SFT (supervised fine-tuning).

### 4.3 3.3 Show Before vs. After Comparison

Demonstrate:

- How the base model responded to instructions before tuning
- How the new *instruct model* responds now

At least **3 comparison examples** must be shown.

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## 5 4. Use a Stronger Pretrained Base Model

Import a stronger BASE model (not instruction-tuned), such as:

- Mistral-7B Base
- Qwen-2 7B Base
- Gemma 7B Base
- Llama-3 8B Base

### 5.1 4.1 Inference Demonstration

Show:

- Text continuation
- Factual Q/A behavior (will often be weak)
- An instruction the model **fails** to follow
  (because base models are not aligned)

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## 6 5. Instruction-Tune the Stronger Base Model

Using LoRA or QLoRA:

### 6.1 5.1 Fine-Tune on an Instruction Dataset

- Show training logs
- Show any metrics available
- Report GPU usage and training speed

### 6.2 5.2 Compare Behavior Before and After

Show at least **3 comparison prompts** demonstrating:

- Before: does not follow instructions
- After: correctly follows instructions

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# 7 6. Reflection

In 5–10 sentences, write about what you learned.
You may discuss:

- How tokenization affected your dataset
- How vocabulary size and context length influence models
- How different sampling methods changed the model's output
- How instruction tuning alters model behavior
- What surprised you about training LLMs
- What you would do differently with more compute
- What you learned about raw base models vs aligned assistants

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# 8 Submission Requirements

You must submit:

1. Your Google Colab notebook(s)
2. A short slide presentation summarizing your findings
3. Screenshots of inference results
4. A short written reflection
5. Your fine-tuned model files (optional if too large)

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# 9 Grading Breakdown

| Criterion | Points |
|---|---|
| Colab setup & GPU explanation | 10 |
| Base model training (dataset + architecture) | 20 |
| Base model inference demos | 10 |
| Instruction tuning of your model | 20 |
| Strong pretrained model inference | 10 |
| Instruction tuning of pretrained model | 20 |
| Reflection & insights | 10 |
| **Total** | **100 points** |

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## 10 End of Assignment

Good luck!
This assignment is hands-on, challenging, and extremely valuable for understanding how real LLMs are developed.

If you have questions, feel free to ask.